

The composition of *shielded payment addresses*, *incoming viewing keys*, *full viewing keys*, and *spending keys* is a cryptographic protocol detail that should not normally be exposed to users. However, user-visible operations should be provided to obtain a *shielded payment address*, *incoming viewing key*, or *full viewing key* from a *spending key* or *extended spending key*.

Users can accept payment from multiple parties with a single *shielded payment address* and the fact that these payments are destined to the same payee is not revealed on the *block chain*, even to the paying parties. *However* if two parties collude to compare a *shielded payment address* they can trivially determine they are the same. In the case that a payee wishes to prevent this they should create a distinct *shielded payment address* for each payer.

[Sapling onward] Sapling and Orchard provide a mechanism to allow the efficient creation of *diversified payment addresses* with the same *spending authority*. A group of such addresses shares the same *full viewing key* and *incoming viewing key*, and so creating as many unlinkable addresses as needed does not increase the cost of scanning the *block chain* for relevant *transactions*.

Note: It is conventional in cryptography to call the key used to encrypt a message in an asymmetric encryption scheme a “*public key*”. However, the *public key* used as the *transmission key* component of an address (pk_{enc} or pk_d) need not be publically distributed; it has the same distribution as the *shielded payment address* itself. As mentioned above, limiting the distribution of the *shielded payment address* is important for some use cases. This also helps to reduce reliance of the overall protocol on the security of the cryptosystem used for note encryption (see § 4.19 ‘*In-band secret distribution (Sprout)*’ on p. 68 and § 4.20 ‘*In-band secret distribution (Sapling and Orchard)*’ on p. 70), since an adversary would have to know pk_{enc} or some pk_d in order to exploit a hypothetical weakness in that cryptosystem.

3.2 Shielded Pools and Notes

A *note* is a representation of value held in a *shielded pool*.

Each *shielded pool* operates according to the rules of a *shielded protocol*, and has its own *treestate* (see § 3.4 ‘*Transactions and Treestates*’ on p. 19). The currently deployed *shielded pools* are the *Sprout pool* and *Sapling pool* and *Orchard pool*.

Each *note* (denoted n) belongs to one of these *shielded pools*. It represents that a value v is spendable by the recipient who holds the *spending key* corresponding to a given *shielded payment address*. The form of a *shielded payment address* is defined by a *shielded protocol*; it can be used to send funds represented by new *notes* to the address, in the *shielded pool(s)* supported by that protocol.

Let MAX_MONEY , ℓ_{PRF}^{Sprout} , $\ell_{PRF}^{Sapling}$, ℓ_d , and ℓ_{value} be as defined in § 5.3 ‘*Constants*’ on p. 77.

Let $NoteCommit^{Sprout}$ be as defined in § 5.4.8.1 ‘*Note Commitments (Sprout)*’ on p. 98.

Let $NoteCommit^{Sapling}$ be as defined in § 5.4.8.2 ‘*Windowed Pedersen commitments*’ on p. 99.

Let $KA^{Sapling}$ be as defined in § 5.4.5.3 ‘*Sapling Key Agreement*’ on p. 92.

Let $DiversifyHash^{Sapling}$ be as defined in § 5.4.1.6 ‘*DiversifyHash^{Sapling} and DiversifyHash^{Orchard} Hash Functions*’ on p. 81.

Let $NoteCommit^{Orchard}$ be as defined in § 5.4.8.4 ‘*Sinsemilla commitments*’ on p. 101.

Let $KA^{Orchard}$ be as defined in § 5.4.5.5 ‘*Orchard Key Agreement*’ on p. 93.

Let $DiversifyHash^{Orchard}$ be as defined in § 5.4.1.6 ‘*DiversifyHash^{Sapling} and DiversifyHash^{Orchard} Hash Functions*’ on p. 81.

Let $PRF^{nfOrchard}$ be as defined in § 5.4.2 ‘*Pseudo Random Functions*’ on p. 89.

Let $q_{\mathbb{P}}$ be as defined in § 5.4.9.6 ‘*Pallas and Vesta*’ on p. 108.